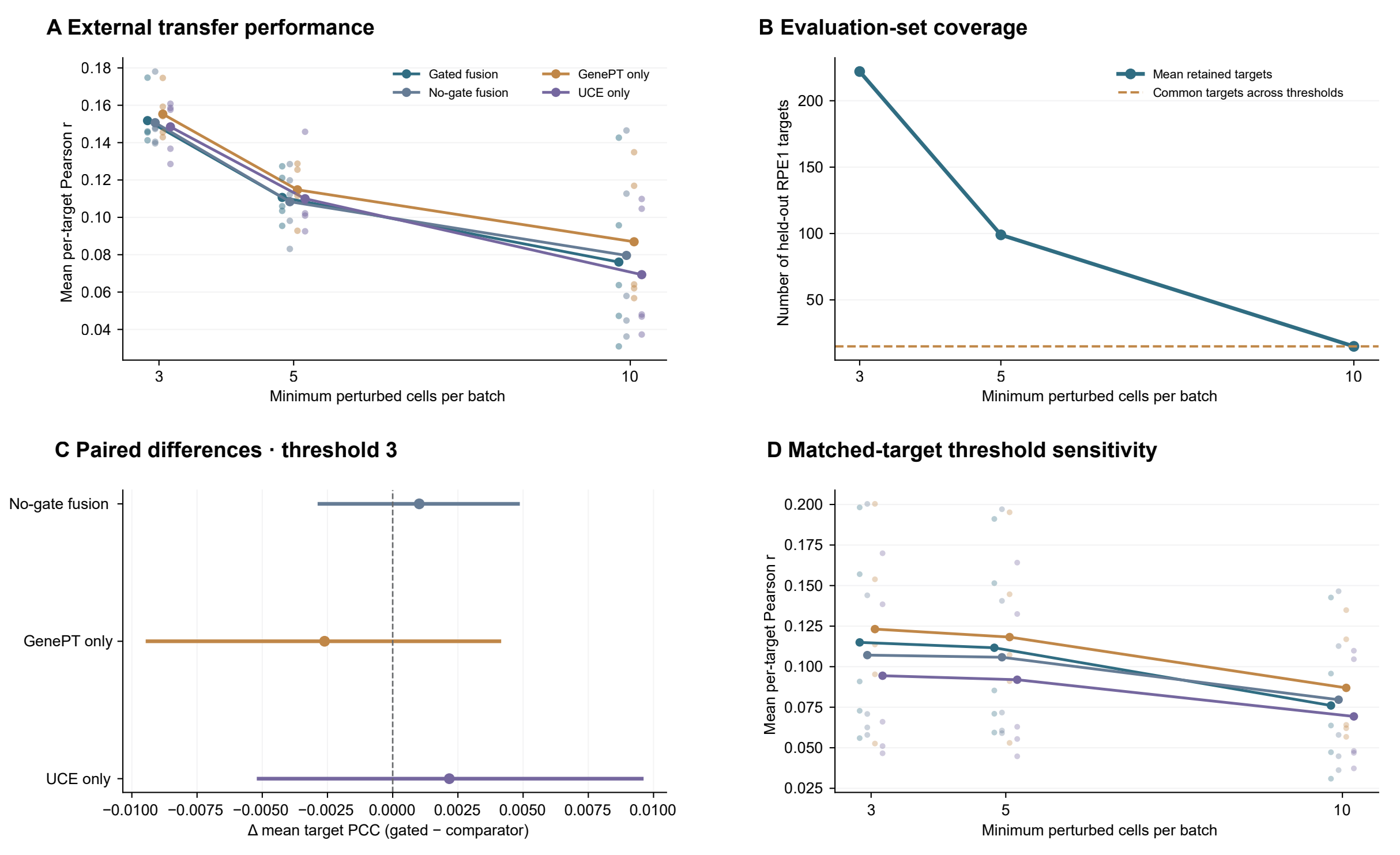

A External transfer performance
Gated fusion
No-gate fusion
GenePT only
UCE only
Mean per-target Pearson r
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
3
5
10
Minimum perturbed cells per batch
B Evaluation-set coverage
Mean retained targets
Common targets across thresholds
Number of held-out RPE1 targets
200
150
100
50
3
5
10
Minimum perturbed cells per batch
C Paired differences · threshold 3
No-gate fusion
GenePT only
UCE only
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
0.0075
0.0100
Δ mean target PCC (gated − comparator)
D Matched-target threshold sensitivity
Mean per-target Pearson r
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
3
5
10
Minimum perturbed cells per batch